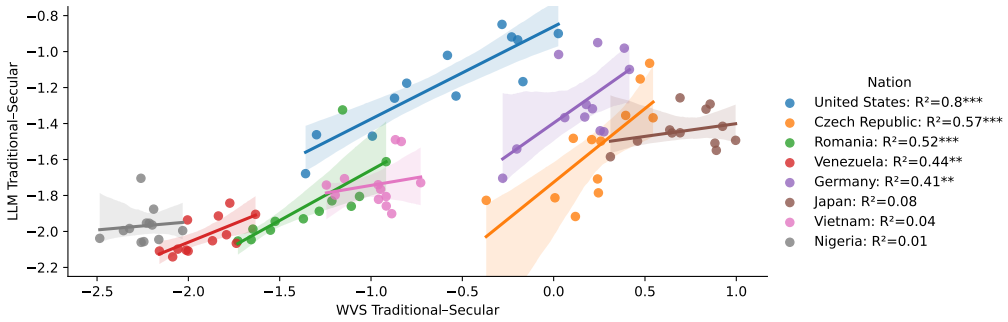

Figure 3: Scatterplot comparing values measured by RVR on LLM-generated text (y-axis) against actual WVS data from humans (x-axis), grouped by country with lines of best fit.

Several crucial points warrant emphasis. First, the United States exhibits the most favorable combination of low error[2] and high explanatory power. Second, Venezuela and Romania display signs of heteroskedasticity as WVS values tend towards secularity, suggesting that LLMs struggle more with modeling increasingly secular values. This indication is further supported by the combination of significant explainable variance but notably higher (and somewhat random) error rates in Germany and especially the Czech Republic, both of which primarily fall above the traditional-secular divide (Figure 3; x=0).

Overall, our results reveal the lowest explanatory power when LLMs generate moral values predictions for non-WEIRD nations (Japan, Vietnam, Nigeria) and the highest error rates when generating predictions for more secular nations (Germany, Czech Republic, Japan) (Table 2). This suggests challenges in capturing belief nuances in both non-WEIRD and more secular nations.

5 RVR Ablation Study: Secular, Traditional, and Combined Values

In this sensitivity analysis, we use only the RVR hypotheses in Table 1 to assess the impact on RVR's moral value scoring. We compare three distinct RVR analyses using (a) only traditional-affirming RVR hypotheses or (b) only secular-affirming hypotheses (i.e., the *inverted* hypotheses in 1, and (c) both traditional and secular hypotheses (i.e., our *combined* set). This approach helps characterize how these hypotheses contribute to latent moral value judgments.

Intuitively, if we only test for resonance with polarized statements such as the extreme-traditional "God is very important in my life" and the extreme-secular "God is irrelevant in my life" then both of these values *conflict* with the moderate statement "God is somewhat important to me." Consequently, we expect that using only extreme-secular hypotheses will incur RVR conflicts that will yield a more *traditional* score, and likewise, using only extreme-traditional hypotheses will incur RVR conflicts that yield a more *secular* score.

Figure 4 shows the results of our ablation study, using only traditional-affirming, only secular-affirming, or the combined categories of RVR hypotheses. These results confirm that using extreme-worded, polarized WVS hypotheses influences the morality judgments in the opposite direction due to the prevalence of conflicts. It also suggests that using RVR with *both* traditional-affirming and secular-affirming hypotheses yields more statistically-bounded results for capturing patterns of relative morality, compared to using polar hypotheses in isolation.

Our results in Figure 4 show that projections calculated using only secular-leaning (Figure 4, 'Polar: Secular') or traditional-leaning (Figure 4, 'Polar: Traditional') hypotheses exhibit significantly higher variance across premises compared to those calculated from the complete set (Figure 4, 'Combined'). This suggests that the moral value projections calculated using the combined set represent the latent moral values in LLM-generated texts with greater accuracy and precision.

[2]Although an RMSE of > 0.5 is not generally considered a good score, we regard it as better than expected considering the data's sparsity and complexity of the task.

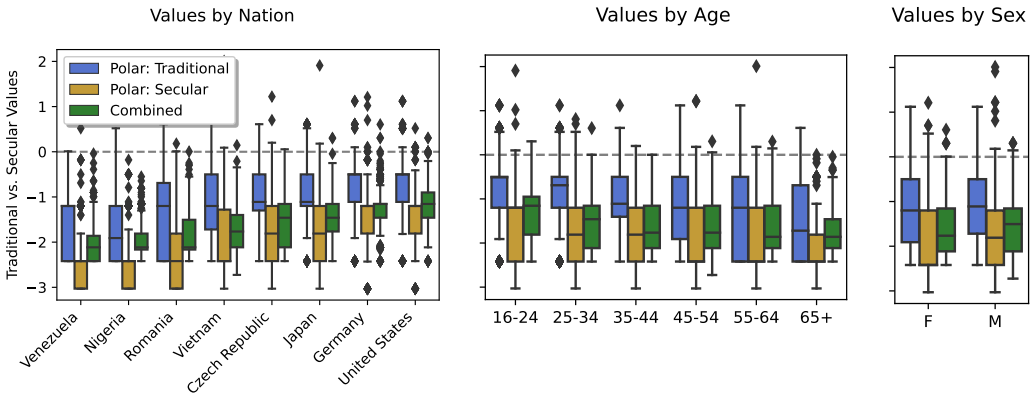

Figure 4: Boxplots comparing traditional-secular ratings by nation, age, and sex, using only traditional-affirming value hypotheses, only secular-affirming value hypotheses, or using the combined (i.e., *complete*) set of hypotheses.

6 Conclusions

This paper presents a novel approach of using Recognizing Value Resonance (RVR) to characterize the implicit moral values within LLM-generated texts. In our analyses of over 50,000 LLM texts, where the LLM writes from the perspectives of various $\langle age, nationality, sex \rangle$ demographics, we use RVR to characterize the LLM's accuracy of assuming various demographic standpoints, using statistics from the World Values Survey (WVS) for comparison. Unlike other approaches that prompt LLMs for agree-or-disagree survey answers, the RVR analysis processes the free-form LLM-generated text, which is also consumed by many end-users.

From a methodological standpoint, there are relatively few methods to characterize implicit moral values in language. Using RVR to anchor language against the WVS is an advance for computational cultural anthropology, as this helps us characterize value plurality and relates moral values to its socio-cultural context. For example, Haslanger wrote that "an adequate account of how implicit bias functions must situate [it] within a broader theory of social structure and structural injustice" [25]. Current LLMs are still lacking dynamically changing moral attitudes situated in diverse social contexts. A misrepresentation of societal values in machine-generated text also poses an ethical concern.

This paper's empirical findings add to the growing support that LLMs have a WEIRD moral bias [26, 9]. More specifically, our results suggest that LLMs may have difficulty assuming non-WEIRD moral perspectives. We also find age inaccuracies in LLM perspective-taking: the LLM expresses more traditional values for older demographics than are measured by the WVS, and tends to over-represent the moral ideals of a younger demographic.

The WEIRD, young over-representation encountered in our results is at odds with the views of Moral Value Pluralism (MVP) [10]. MVP advocates the inclusion of diverse, potentially-conflicting values of diverse cultures rather than favoring dominant cultures' values or reducing diverse moral values into a consistent subset. This makes it an ideal tool for investigating moral value alignment in AI systems. If a LLM were consistent with Moral Value Pluralism, prompting it to assume diverse cultural or demographic perspectives would result in an accurate expression of implicit and explicit cultural and moral values. At present, we have evidence that LLMs are biased to assume a culturally-dominant WEIRD voice rather than accurately expressing a plurality of cultural voices.

We hypothesize that the patterns of implicit moral values we observe in LLMs result from the largely Western and English nature of the training data [13, 27]. Under-representation of text generated by older adults might also contribute to these patterns, thus the model may be echoing the values of people who contribute most to Internet text. These LLM biases may be difficult to correct unless new corpora are used that explicitly include older adults and under-represented cultures. Moral value inaccuracies for women and non-WEIRD countries reflects other common biases of LLMs and is not surprising given previous literature [9, 10]. To our knowledge this is the first quantification of implicit value bias using a validated survey, the WVS, as a control.

6.1 Limitations and Future Work

There are many nuances around LLM usage, prompt design, value alignment and value plurality that we would like to address in future works that were not part of this work. Firstly, it would be interesting to test and compare a variety of LLMs, since previous works [28] have shown subtle differences in the ways different LLMs write responses, and the biases embedded in these responses. Additionally, many interesting questions remain regarding prompt designs. We opted to ask questions to LLMs in English about what a person of a given nationality would answer, but a valid alternative may be asking translations of the same question to LLMs. Initial experiments with translations yielded extremely similar results and/or the models refusing to answer meaningfully to avoid stereotyping, motivating more work on prompt design. Further, it would be interesting to probe the role of other demographics (such as social class or urban/suburban/rural) or other identities (going beyond male/female gender identity). Additionally, while this work asked LLMs to predict responses to an ethnographic survey, further work could probe the difference between LLMs' responses when asked to provide survey responses versus what a person of a given demographic group would think. We opted to ask for survey responses for a fairer comparison to WVS results. Finally, the choice of LLM parameters could be varied in the future to provide a more comprehensive study of how the degree of determinism or creativity in the LLM affects biases. Motoki et al. [29] and others address similar concerns in their work.

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